Functional Requirements: Smart Curb

1. User Accounts and Roles:

The system shall install an occupancy sensor on every curb space.

The system shall detect when a vehicle arrives and leaves, and record the exact time of each

The system send sensor readings to the platform in real time, so zone availability is always up to date.

The system link each sensor to one curb space and one zone, so a reading always belongs to the right place.

The system combine the sensor occupancy reading with the plate recognition camera reading so each vehicle is tied to a specific space and time.

The system start the plate recognition camera when a sensor detects a new vehicle so the plate is captured at the moment of arrival

The system detect when a space is occupied with no booking and report it as a possible violation.

The system shall detect when a vehicle stays after its booking ends and start the 5-minute violation timer from the sensor departure data.

The system shall free a booked slot for others if the vehicle never arrives within a set grace period based on sensor data.

Sensor Management:

- The system shall monitor the health of every sensor and camera (online, offline or faulty).
- The system alert the technical staff when a sensor or camera stops working or sends strange readings.
- The system shall let admins register, configure and calibrate sensors from a dashboard.
- The system mark a space as “unverified” when its sensor is offline and shall not issue automatic fines from that space until it is checked.
- The system shall keep all sensor data for reports and as part of the violation evidence.

Updates to Earlier Requirements

- so it can be removed or merged.
- now use sensor data together with the camera
- should include the sensor reading and its timestamp together with the plate photo because this makes the fine stronger legally.
- should use sensor data for occupancy and stop duration.

The Traffic Police need legally valid fines so the system should not fine anyone when the sensor may be broken. That is why I added the unverifie rule. It is my own addition so you can remove it if your doctor or team doesn’t want it.